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Subject Code/Name: MLA0305-Reinforcement Learning

1. Aim

To install and configure the Anaconda Navigator, Python, TensorFlow, Keras, and Gymnasium software environment required for implementing and experimenting with Reinforcement Learning algorithms.

2. Procedure

Download and install Anaconda Distribution from the official Anaconda website.

Open Anaconda Navigator after completing the installation.

Create a separate Python environment for Reinforcement Learning experiments to avoid package conflicts.

Select an appropriate Python version while creating the environment.

Launch Jupyter Notebook from Anaconda Navigator.

Install and configure the required Reinforcement Learning libraries, including:

NumPy

Matplotlib

TensorFlow

Keras

Gymnasium

Configure TensorFlow for machine-learning and deep-learning operations.

Configure Keras as the high-level deep-learning API used with TensorFlow.

Configure Gymnasium to provide standard Reinforcement Learning environments such as CartPole, MountainCar, and FrozenLake.

Verify that all required software packages are properly installed and accessible within the created Python environment.

Open Jupyter Notebook and ensure that the configured Python environment is available as a notebook kernel.

The configured environment can then be used for implementing various Reinforcement Learning algorithms in the subsequent experiments.

3. Software / Tools Required

S. No. Software / Tool Purpose

- | | | |
|---|--------------------|--|
| 1 | Anaconda Navigator | Environment and package management |
| 2 | Python | Programming language |
| 3 | Jupyter Notebook | Writing and executing Python programs |
| 4 | NumPy | Numerical and mathematical operations |
| 5 | Matplotlib | Data visualization |
| 6 | TensorFlow | Machine Learning and Deep Learning framework |
| 7 | Keras | High-level Deep Learning API |
| 8 | Gymnasium | Reinforcement Learning environments |

4. Visualization

The successful configuration can be visually represented through the Anaconda Navigator environment, showing the created Python environment and installed packages.

The Jupyter Notebook interface can also be used to verify that the configured Python environment is available and ready for Reinforcement Learning experiments.

Expected visualization:

Anaconda Navigator → Python Environment → Jupyter Notebook → TensorFlow / Keras / Gymnasium

5. Summary

In this experiment, an environment required for Reinforcement Learning experimentation was prepared using Anaconda Navigator and Python. The necessary libraries such as TensorFlow, Keras, Gymnasium, NumPy, and Matplotlib were configured. Jupyter Notebook was used as the development environment for executing Python-based Reinforcement Learning experiments.

6. Result

Thus, Anaconda Navigator, Python, TensorFlow, Keras, and Gymnasium were successfully installed and configured, providing a suitable environment for implementing and analyzing Reinforcement Learning algorithms.